



MOHAMMAD ROHAAN

BSCS Candidate · FAST NUCES, Islamabad

Software Engineering · AI/ML · Data Science

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OBJECTIVE

BSCS candidate at **FAST NUCES, Islamabad** (expected graduation Jan 2027). Delivered end-to-end projects in **AI/ML** (live bilingual voice agent, NLP quiz generation, YOLOv8 detection), software engineering (**Spring Boot** library platform with Scrum and full test coverage), database systems (SQL Server + PHP hospital application), and systems programming (OpenMP K-means). Seeking software engineering or AI/ML internships and full-time roles where I can ship reliable, measurable work.

EDUCATION

FAST NUCES — Islamabad, Pakistan

Expected Jan 2027

Bachelor of Science in Computer Science (BSCS)

Final-year BSCS candidate

TECHNICAL SKILLS

Languages: Python, Java, C, C++, C#, JavaScript, SQL, PHP

AI / ML / Data: scikit-learn, pandas, NLP (TF-IDF, BLEU/ROUGE/METEOR), YOLOv8, Streamlit, VAPI / Groq voice AI, model evaluation, feature engineering

Software Eng: Java 17, Spring Boot 3.5, Spring MVC, Thymeleaf, Spring Security, Spring Data JPA, Hibernate, MySQL 8, Next.js, Scrum / Agile, REST APIs, Git, Maven

Web & Databases: HTML, CSS, JavaScript, PHP, SQL Server, T-SQL, ER / EER modeling, normalization

Systems: OpenMP, parallel computing, multithreading, benchmarking, DSA, Linux

Embedded / Robotics: Arduino, ESP32, Raspberry Pi, ultrasonic sensors, MATLAB (robotics simulation basics)

RELEVANT COURSEWORK

Data Structures & Algorithms, Operating Systems, Computer Networks, Database Systems, Software Engineering, Probability & Statistics, Parallel & Distributed Computing, Numerical Computing / HPC, Design & Analysis of Algorithms, Machine Learning / Data Science, Computer Vision.

SELECTED PROJECTS

AI Voice Cold-Calling Agent (Next.js, VAPI, Groq, Deepgram) · **Live** · **GitHub**

- Browser and phone voice agent with English/Urdu, objection handling, recordings, transcripts, and call history.
- Deployed on Vercel; STT → LLM → TTS pipeline with outbound dialer and call-history UI.

AI Quiz Generator (Python, Streamlit, scikit-learn) · **GitHub**

- Dual ML pipelines on RACE for question/answer ranking, distractors, and graduated hints; Streamlit quiz UI.
- Evaluated with BLEU, ROUGE, and METEOR (hint METEOR 0.69).

LibraryMS (Java 17, Spring Boot, MySQL, Scrum) · **GitHub**

- Role-based library platform: catalog, borrow/return, FIFO reservations, fines, and admin reports.
- Delivered in Scrum sprints; 22/22 black-box and white-box tests passed.

Shuttlecock Detection (YOLOv8, transfer learning) · **GitHub**

- Fine-tuned detector on 15k labeled images for robotic pickup; precision, recall, and mAP.
- Real-time camera inference toward a Raspberry Pi + Arduino collection stack.

Hospital DB System (SQL Server, PHP, ER/EER) · **GitHub**

- Normalized hospital schema with integrity constraints and 12 operational SQL queries.
- PHP/JS portals for intake, ward admin, and staff reports.

K-means Triangle Inequality (OpenMP) — Elkan vs Lloyd scheduling and dimension benchmarks. **GitHub**

Git Lite (C++) — Mini Git with commit DAG, hashing, and diff/merge. **GitHub**